The isoperimetric inequality

Andrea Martelli

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Abstract

These notes are the content of my seminar for the final exam of the course *Advanced Topics in Analysis*, taught by Prof. Gian Paolo Leonardi during the Fall semester of 2025 at University of Trento.

1 Introduction

In these notes we are going to prove the isoperimetric inequality, characterizing the equality case. The main reference is [Mag12, §13.2 and Chapter 14].

Theorem 1.1 (Isoperimetric inequality)**.** *For every Lebesgue measurable set $E \subset \mathbb{R}^n$,*

$$P(E) \geq n\omega_n^{1/n}|E|^{\frac{n-1}{n}},$$

where $\omega_n = |B_1|$ is the n-volume of the standard unit ball of $\mathbb{R}^n$. Moreover, the equality holds if and only if E is equivalent to a ball, meaning that there exist $x \in \mathbb{R}^n$ and $R > 0$ (that necessarily satisfies $|E| = \omega_n R^n$) such that $|E \triangle B_R(x)| = 0$.

In other words, for every $R > 0$

$$\inf\{P(E) \colon E \subset \mathbb{R}^n, |E| = \omega_n R^n\} = n\omega_n R^{n-1}$$

and the infimum is realized only by sets equivalent to balls of radius R.

When $n = 1$, this is quite easy.

Theorem 1.2 (1-dimensional isoperimetric inequality)**.** *If $E \subset \mathbb{R}$ is a Lebesgue measurable set with $|E| < \infty$, then*

$$P(E) \geq 2,$$

with equality if and only if E is equivalent to an interval of length $|E|$.

Proof. Assume that $P(E) < \infty$. Then E is equivalent to a finite union of disjoint open intervals, so $P(E) = 2N$ where N is the number of intervals. $\square$

For $n > 1$, the proof of the inequality is not hard: on bounded sets it's just the direct method (cf. Lemma 4.1), and the unbounded case can be obtained by approximating finite perimeter sets with bounded finite perimeter sets.

The actual work is to characterize the equality, i.e., minimizers of the variational problem

$$\inf\{P(E)\colon E \subset \mathbb{R}^n \text{ Lebesgue measurable}, |E| = m\}, \tag{1}$$

for $m > 0$ fixed. A way to proceed is by using the Steiner inequality (cf. Theorem 3.2), which states that the Steiner symmetrization does not increase the perimeter. This implies that minimizers must have many symmetries, and it's not hard to deduce that they are balls. Therefore, a sketch of the proof is to show, in order

- finite perimeter sets can be approximated by bounded sets with polyhedral boundary (cf. Theorem 2.4);
- the Steiner inequality, proving it first on sets with polyhedral boundary and then approximating;
- minimizers of (1) must be equivalent to balls, using the Steiner inequality.

Another approach that I find interesting is considering the problem from the point of view of differential geometry. Consider the class of smooth, connected, compact hypersurfaces that are the boundary of some open bounded set Ω. The critical points of the $(n-1)$-area functional (i.e., the $(n-1)$-Hausdorff measure restricted to the class of such hypersurfaces) with respect to volume preserving variations are called *CMC* (because, by the first variation formula, they have constant mean curvature). Therefore, any minimizer of the variational problem

$$\inf\left\{\mathscr{H}^{n-1}(\Sigma) \,\middle|\, \begin{array}{l} \Sigma = \partial\Omega \text{ smooth, connected,} \\ \text{compact hypersurface}, |\Omega| = m \end{array}\right\}.$$

(for $m > 0$ fixed) must be CMC and must be stable with respect to volume preserving variations. However, Barbosa and do Carmo ([BC84], cf. Theorem 5.1) proved that the only smooth, connected, compact hypersurfaces of $\mathbb{R}^n$ are the round spheres. The proof, sketched in the last section of these notes, is quite simple and exploits the stability inequality to prove that every CMC-stable hypersurface must be totally umbilical, hence, if it's connected, a sphere.

2 Polyhedral approximation

Given a function $u\colon \mathbb{R}^{n-1} \to \mathbb{R}$ and a subset $G \subset \mathbb{R}^{n-1}$, we define the *graph of u over G* as

$$\Gamma(u, G) \coloneqq \{(x, u(x)) \in \mathbb{R}^n \colon x \in G\}.$$

Definition 2.1. A *polyhedral set* is a finite, non empty intersection of closed half-spaces.

A continuous function $u\colon G \subset \mathbb{R}^n \to \mathbb{R}$ is said to be *finitely piecewise affine* if there exists a finite partition of polyhedral sets $\{G_i\}_{i=1}^k$ of G such that u is affine on each G_i.

A set $E \subset \mathbb{R}^n$ is said to have *polyhedral boundary* if for every $x \in \partial E$ there exist $r > 0$, a finitely piecewise affine function $u\colon (-r,r)^{n-1} \to \mathbb{R}$ and an isometry $T\colon \mathbb{R}^n \to \mathbb{R}^n$ with $T(x) = 0$ such that

$$\begin{aligned} T(\partial E) \cap (-r,r)^n &= G(u, (-r,r)^{n-1}) \\ T(E) \cap (-r,r)^n &= \{(z,t) \in (-r,r)^{n-1} \times (-r,r) \colon t > u(z)\}. \end{aligned}$$

Remark 2.2. It's not hard to prove that any function $u \in C_c^1(\mathbb{R}^n)$ can be approximated by compactly supported finitely piecewise affine functions in the C^1-norm.

Lemma 2.3 (Approximation by finitely piecewise affine functions)**.** *For every $u \in BV(\mathbb{R}^n)$ there exists a sequence $u_h\colon \mathbb{R}^n \to \mathbb{R}$ of finitely piecewise affine functions such that $u_h \to u$ in the BV-strict sense, that is, $u_h \to u$ in $L^1(\mathbb{R}^n)$ and*

$$\int_{\mathbb{R}^n} |\nabla u_h| \mathrm{d}z \to |Du|(\mathbb{R}^n).$$

Proof. Step 1. Assume that $u \in C^1(\mathbb{R}^n) \cap BV(\mathbb{R}^n)$. Take a sequence of cutoff functions $\varphi_h\colon \mathbb{R}^n \to \mathbb{R}$, that is, for some sequence $R_h \nearrow \infty$, smooth functions with compact support in $B_{R_h+1/h}$, $0 \le \varphi \le 1$ on $\mathbb{R}^n$, $\varphi_h = 1$ on B_{R_h} and $\|\nabla \varphi_h\|_{C^0} \le C$, $C > 0$ not depending on h. Then $v_h := u\varphi_h \in C_c^1(\mathbb{R}^n)$, $v_h \to u$ in $L^1(\mathbb{R}^n)$, and

$$\begin{aligned} \int_{\mathbb{R}^n} |\nabla v_h| \mathrm{d}x &= \int_{B_{R_h}} |\nabla u| \mathrm{d}x + \int_{B_{R_h+1/h} \setminus B_{R_h}} |\nabla v_h| \mathrm{d}x \\ &\to \int_{\mathbb{R}^n} |\nabla u| \mathrm{d}x \qquad \text{as } h \to \infty. \end{aligned}$$

For every $h \in \mathbb{N}$, take a sequence $v_h^k\colon \mathbb{R}^n \to \mathbb{R}$ of compactly supported finitely piecewise affine functions approximating each v_h in $C_c^1(\mathbb{R}^n)$ with respect to the C^1-norm. Then in particular $v_h^k \to v_h$ in $L^1(\mathbb{R}^n)$ and

$$\lim_{k\to\infty} \int_{\mathbb{R}^n} |\nabla v_h^k| \mathrm{d}x = \int_{\mathbb{R}^n} |\nabla v_h| \mathrm{d}x.$$

With a diagonal argument, we can extract a diagonal sequence $u_h := v_h^{k(h)}$ that satisfies the required properties.

Step 2. Now take any $u \in BV(\mathbb{R}^n)$, and fix a kernel mollifier $\rho \in C_c^\infty(\mathbb{R}^n)$. Let $v_h = u \star \rho_{\varepsilon_h}$, with $\varepsilon_h \to 0$, and for each $h \in \mathbb{N}$ take a sequence $v_h^k\colon \mathbb{R}^n \to \mathbb{R}$ of finitely piecewise affine functions approximating each v_h, i.e., such that $v_h^k \to v_h$ in $L^1(\mathbb{R}^n)$ and

$$\lim_{k\to\infty} \int_{\mathbb{R}^n} |\nabla v_h^k| \mathrm{d}x = \int_{\mathbb{R}^n} |\nabla v_h| \mathrm{d}x.$$

By the properties of mollification,

$$\lim_{h\to\infty}\int_{\mathbb{R}^n}|\nabla u_h|\mathrm{d}x = |D\chi_E|(\mathbb{R}^n) = P(E),$$

so, as before, we can extract a diagonal sequence $u_h := v_h^{k(h)}$ that satisfies the required properties. □

Theorem 2.4 (Polyhedral approximation)**.** *Let $E \subset \mathbb{R}^n$ be a set with $|E| < \infty$ and $P(E) < \infty$. Then there exists a sequence of bounded sets $E_h \subset \mathbb{R}^n$ with polyhedral boundary such that*

$$E_h \to E, \qquad P(E_h) \to P(E).$$

Remark 2.5. In the proof, we are going to use the following formula: for every measurable set $E \subset \mathbb{R}^n$ and $u, v \in L^1_{loc}(\mathbb{R}^n)$,

$$\int_E |u-v|\mathrm{d}x = \int_{\mathbb{R}} \mathscr{L}^n(E \cap \{u > t\} \triangle \{v > t\})\mathrm{d}t, \tag{2}$$

where $A \triangle B = (A \setminus B) \cup (B \setminus A)$ is the symmetric difference. This is follows from the usual interpretation of $\int_E |u-v|$ as the area between the graph of u and the graph of v over E. Indeed, consider separately $E^+ = E \cap \{u > v\}$ and $E^- = E \cap \{u < v\}$. By Fubini's theorem,

$$\begin{aligned}
\int_{E^+} |u-v|\mathrm{d}x &= \int_{E^+} \mathscr{L}^1((v(x), u(x)))\mathrm{d}x \\
&= \mathscr{L}^{n+1}(\{(x,t) \in E \times \mathbb{R} \colon v(x) < t < u(x)\}) \\
&= \int_{\mathbb{R}} \mathscr{L}^n(\{x \in E \colon v(x) < t < u(x)\})\mathrm{d}t \\
&= \int_{\mathbb{R}} \mathscr{L}^n(E \cap \{u > t\} \setminus \{v > t\})\mathrm{d}t
\end{aligned}$$

and similarly for E^-.

We are going to use the following lemma, which is independent from all the results given in these notes.

Lemma 2.6 ([Mag12, Lemma 15.2])**.** *Let $E \subset \mathbb{R}^n$ be a set of locally finite perimeter and let $x \in \mathbb{R}^n$. Then for every $R > 0$ the set $E \cap B_R(x)$ has finite perimeter in $\mathbb{R}^n$. Moreover, for a.e. $R > 0$*

$$P(E \cap B_R(x)) = P(E; B_R(x)) + \mathscr{H}^{n-1}(E \cap \partial B_R(x)).$$

Proof of Theorem 2.4. Step 1. By Lemma 2.3, there is a sequence of piecewise affine functions $u_h \colon \mathbb{R}^n \to \mathbb{R}$ such that $u_h \to \chi_E$ in $L^1(\mathbb{R}^n)$ and

$$\lim_{h\to\infty}\int_{\mathbb{R}^n}|\nabla u_h|\mathrm{d}x = P(E). \tag{3}$$

For every $h \in \mathbb{N}$ and $t \in (0,1)$ consider the set

$$E_h^t := \{u_h > t\}.$$

Since u_h is piecewise affine, $\partial E_h^t \subset \{u_h = t\}$ is always a polyhedra. We are going to choose a suitable $t \in (0,1)$.

By (2) we have that

$$\begin{aligned}
\int_0^1 \int_{\mathbb{R}^n} |\chi_{E_h^t} - \chi_E| \mathrm{d}x \mathrm{d}t &= \int_0^1 \int_{\mathbb{R}^n} |\chi_{\{u_h > t\}} - \chi_{\{\chi_E > t\}}| \mathrm{d}x \mathrm{d}t \\
&= \int_0^1 \mathscr{L}^n(\{u_h > t\} \triangle \{\chi_E > t\}) \mathrm{d}t \\
&\leq \int_{\mathbb{R}} \mathscr{L}^n(B \cap \{u_h > t\} \triangle \{\chi_E > t\}) \mathrm{d}t \\
&= \int_{\mathbb{R}^n} |u_h - \chi_E| \mathrm{d}x \to 0,
\end{aligned}$$

so $E_h^t \to E$ for a.e. $t \in (0,1)$, and in particular

$$P(E) \leq \liminf_{h \to \infty} P(E_h^t) \text{ for a.e. } t \in (0,1) \tag{4}$$

by the lower semicontinuity of the perimeter. By (3),

$$\begin{aligned}
P(E) &= \lim_{h \to \infty} \int_{\mathbb{R}^n} |\nabla u_h| \mathrm{d}x \\
&= \lim_{h \to \infty} \int_{\mathbb{R}} P(\{u_h > t\}) \mathrm{d}t \qquad \text{(coarea formula)} \\
&\geq \int_{\mathbb{R}} \liminf_{h \to \infty} P(\{u_h > t\}) \mathrm{d}t \quad \text{(Fatou's lemma)} \\
&\geq \int_0^1 \liminf_{h \to \infty} P(E_h^t) \mathrm{d}t \\
&\geq P(E),
\end{aligned}$$

having used (4) in the last inequality. Therefore,

$$P(E) = \int_0^1 \liminf_{h \to \infty} P(E_h^t) \mathrm{d}t,$$

and again by (4) it must be

$$P(E) = \liminf_{h \to \infty} P(E_h^t) \qquad \text{for a.e. } t \in (0,1).$$

Up to passing to a subsequence (which we don't relabel),

$$P(E) = \lim_{h \to \infty} P(E_h^t) \qquad \text{for a.e. } t \in (0,1). \tag{5}$$

Thus we can find $t \in (0,1)$ such that $E_h := E_h^t$ is as required.

Step 2. Now suppose that E is not bounded. We will find a sequence $R_i \nearrow \infty$ such that $E \cap B_{R_i} \to E$ and $P(E \cap B_{R_i}) \to P(E)$. Then we just need to approximate each $E \cap B_{R_i}$ using the previous step. Since $|E| < \infty$, $E \cap B_{R_i} \to E$ and whenever $R_i \to \infty$, so we have to choose R_i only worrying to achieve $P(E \cap B_{R_i}) \to P(E)$. First notice that, since $P(E) < \infty$ (i.e. $|D\chi_E|$ is a finite measure), whenever $R_i \nearrow \infty$ we have

$$\lim_{i\to\infty} P(E; \mathbb{R}^n \setminus B_{R_i}) = \lim_{i\to\infty} |D\chi_E|(\mathbb{R}^n \setminus B_{R_i}) = 0$$
$$\lim_{i\to\infty} P(E; B_{R_i}) = \lim_{i\to\infty} |D\chi_E|(B_{R_i}) = P(E)$$

by continuity from above and below respectively. By Lemma 2.6,

$$P(E \cap B_R) = P(E; B_R) + \mathscr{H}^{n-1}(E \cap \partial B_R) \quad \text{for a.e. } R > 0. \tag{6}$$

By the coarea formula applied to the function $\rho(x) = |x|$,

$$\int_0^\infty \mathscr{H}^{n-1}(E \cap \partial B_R)\mathrm{d}R = \int_E |\nabla\rho|\mathrm{d}x = |E| < \infty$$

since $\nabla\rho(x) = x/|x|$. Therefore, there exists a sequence $R_i \nearrow \infty$ such that (6) holds for every R_i and

$$\mathscr{H}^{n-1}(E \cap \partial B_{R_i}) \to 0.$$

Hence,

$$\begin{aligned}|P(E) - P(E \cap B_{R_i})| &\le |P(E) - P(E; B_{R_i})| + |P(E; B_{R_i}) - P(E \cap B_{R_i})| \\ &= |P(E) - P(E; B_{R_i})| + \mathscr{H}^{n-1}(E \cap \partial B_{R_i}) \to 0.\end{aligned}$$

as $i \to \infty$, as required. □

3 Steiner inequality

Let us recall the definition of Steiner symmetrization. Write $\mathbb{R}^n = \mathbb{R}^{n-1} \times \mathbb{R}$ with coordinates $x = (z,t)$, $z \in \mathbb{R}^{n-1}$, $t = x_n \in \mathbb{R}$. For every $E \subset \mathbb{R}^n$ and $z \in \mathbb{R}^{n-1}$, consider the *vertical z-slice*

$$E_z := \{t \in \mathbb{R} \colon (z,t) \in E\} \subset \mathbb{R}.$$

The *Steiner symmetrization of E with respect to the direction e_n* is the set

$$S_{e_n}(E) := \left\{(z,t) \in \mathbb{R}^{n-1} \times \mathbb{R} \colon |t| < \frac{1}{2}\mathscr{L}^1(E_z)\right\}.$$

By Fubini's theorem, $|S_{e_n}(E)| = |E|$. It is also quite easy to prove that

$$\operatorname{diam}(S_{e_n}(E)) \le \operatorname{diam}(E).$$

If $v \in \mathbb{R}^n$ is a unit vector, the *Steiner symmetrization of E with respect to the direction v* is obtained by choosing any orthogonal change of coordinate $T\colon \mathbb{R}^n \to \mathbb{R}^n$ such that $T(v) = e_n$ and letting

$$S_v(E) \coloneqq T^{-1}(S_{e_n}(T(E))).$$

Since T is an isometry, $|S_v(E)| = |E|$ and $\operatorname{diam}(S_v(E)) \leq \operatorname{diam}(E)$.

Proposition 3.1 (Slicing perimeter by lines)**.** *Let $E \subset \mathbb{R}^n$ be a Lebesgue measurable set with $|E| < \infty$ and $P(E) < \infty$. Then,*

$$\int_{\mathbb{R}^{n-1}} P(E_z)\mathrm{d}z \leq P(E).$$

and in particular the vertical z-slice E_z is a set of finite perimeter in $\mathbb{R}$ for a.e. $z \in \mathbb{R}^{n-1}$.

Proof. Fix a mollifier $\rho \in C_c^\infty(B_1)$ and let $u_\varepsilon \coloneqq \chi_E \star \rho_\varepsilon$, so that $u_\varepsilon \to \chi_E$ and

$$P(E) = \lim_{\varepsilon \to 0} \int_{\mathbb{R}^n} |\nabla u_\varepsilon| \mathrm{d}x. \tag{7}$$

By Fubini's theorem,

$$\int_{\mathbb{R}^{n-1}} \int_{\mathbb{R}} |u_\varepsilon(z,t) - \chi_{E_z}(t)| \mathrm{d}t\mathrm{d}z = \int_{\mathbb{R}^n} |u_\varepsilon - \chi_E| \mathrm{d}x \to 0$$

since $u_\varepsilon \to \chi_E$ in L^1, so there exists a sequence $\varepsilon_h \searrow 0$ such that

$$u_h(z,\cdot) \coloneqq u_{\varepsilon_h}(z,\cdot) \to \chi_{E_z} \qquad \text{for a.e. } z \in \mathbb{R}^{n-1}.$$

By the lower semicontinuity of the total variation,

$$P(E_z) \leq \liminf_{h\to\infty} \int_{\mathbb{R}} |\partial_t u_h(z,t)| \,\mathrm{d}t \leq \liminf_{h\to\infty} |\nabla u_h(z,t)| \mathrm{d}t,$$

and by Fatou's lemma, Fubini's theorem and (7), we get

$$\begin{aligned}
\int_{\mathbb{R}^{n-1}} P(E_z)\mathrm{d}z &\leq \int_{\mathbb{R}^{n-1}} \left(\liminf_{h\to\infty} \int_{\mathbb{R}} |\nabla u_h(z,t)| \mathrm{d}t \right) \mathrm{d}z \\
&\leq \liminf_{h\to\infty} \int_{\mathbb{R}^{n-1}} \int_{\mathbb{R}} |\nabla u_h(z,t)| \mathrm{d}t\mathrm{d}z \\
&= \lim_{h\to\infty} |\nabla u_h| \mathrm{d}x = P(E),
\end{aligned}$$

as required. □

Theorem 3.2 (Steiner inequality)**.** *Let $E \subset \mathbb{R}^n$ be a set with $|E| < \infty$ and finite perimeter $P(E) < \infty$. Then for every unit vector $v \in \mathbb{R}^n$*

$$P(S_v(E)) \leq P(E),$$

and the equality holds if and only the vertical z-slice E_z is an interval for a.e. $z \in \mathbb{R}^{n-1}$.

In particular, if E is equivalent to a convex set, then the equality holds if and only if E is equivalent to $S_v(E) + cv$ for some $c \in \mathbb{R}$.

Proof. Without loss of generality, assume that $v = e_n$ and denote $E^s \coloneqq S_{e_n}(E)$. We divide the proof in three steps.

Step 1. Assume that E is a bounded set with polyhedral boundary ∂E, and that the outer unit normal to E (defined $\mathscr{H}^{n-1}$-a.e. on ∂E) is never orthogonal to e_n. In particular, $P(E) < \infty$ and $|D\chi_E| = \mathscr{H}^{n-1} \llcorner \partial E$. Let G be the projection of E on $\mathbb{R}^{n-1}$, or equivalently

$$G = \{z \in \mathbb{R}^{n-1} \colon \mathscr{L}^1(E_z) > 0\}.$$

Then in the definition of set with polyhedral boundary, we can always take T to be a translation, so there exist a finite partition $\{G_h\}_{h=1}^M$ of G made of $(n-1)$-polyhedral sets and affine functions

$$v_h^1, u_h^1, \dots, v_h^{N(h)}, u_h^{N(h)} \colon \mathbb{R}^{n-1} \to \mathbb{R},$$

such that

$$\partial E = \bigcup_{h=1}^{M} \bigcup_{k=1}^{N(h)} \Gamma(u_h^k, G_h) \cup \Gamma(v_h^k, G_h)$$
$$E = \bigcup_{h=1}^{M} \left\{ (z,t) \in G_h \times \mathbb{R} \colon t \in \bigcup_{k=1}^{N(h)} (v_h^k(z), u_h^k(z)) \right\}.$$

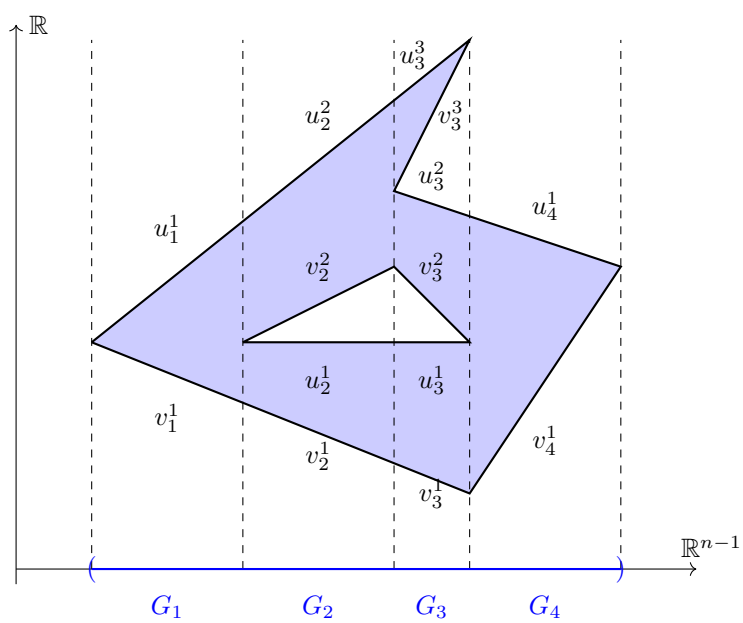

Figure 1: Notation in step 1 of the proof of Theorem 3.2.

With this notation,

$$\begin{aligned} P(E) = \mathscr{H}^{n-1}(\partial E) &= \sum_{h=1}^{M} \mathscr{H}^{n-1}(\partial E \cap G_h \times \mathbb{R}) \\ &= \sum_{h=1}^{M} \int_{G_h} \sum_{k=1}^{N(h)} \left(\sqrt{1+|\nabla v_h^k|^2} + \sqrt{1+|\nabla u_h^k|^2} \right) \mathrm{d}z. \end{aligned} \tag{8}$$

For every $z \in \mathbb{R}^{n-1}$,

$$E_z = \begin{cases} \bigcup_{k=1}^{N(h)} (v_h^k(z), u_h^k(z)) & \text{if } z \in G_h, \\ \varnothing & \text{if } z \in \mathbb{R}^{n-1} \setminus G \end{cases}$$

and, if we set $m(z) = \mathscr{L}^1(E_z)$ for every $z \in \mathbb{R}^{n-1}$,

$$m(z) \coloneqq \mathscr{L}^1(E_z) = \begin{cases} \sum_{k=1}^{N(h)} u_h^k(z) - v_h^k(z) & \text{if } z \in G_h, \\ 0 & \text{if } z \in \mathbb{R}^{n-1} \setminus G. \end{cases} \tag{9}$$

Observe that $m \colon \mathbb{R}^{n-1} \to \mathbb{R}$ is affine on each G_h. Therefore,

$$E^s = \{(z,t) \in \mathbb{R}^{n-1} \times \mathbb{R} \colon |t| < m(z)/2\}$$

is a bounded set with polyhedral boundary, so in particular it has finite perimeter and $|D\chi_{E^s}| = \mathscr{H}^{n-1} \llcorner \partial E^s$. By the area formula,

$$\begin{aligned} P(E^s) = \mathscr{H}^{n-1}(\partial E^s) &= 2 \int_G \sqrt{1 + \left|\frac{1}{2}\nabla m\right|^2} \mathrm{d}z \\ &= \sum_{h=1}^{M} \int_{G_h} \sqrt{4 + |\nabla m|^2} \mathrm{d}z \end{aligned} \tag{10}$$

By (9) and the convexity of the function $z \mapsto \sqrt{1+|z|^2}$, for every $h = 1, \dots, M$

$$\begin{aligned} \sum_{k=1}^{N(h)} \sqrt{1+|\nabla v_h^k|^2} + \sqrt{1+|\nabla u_h^k|^2} &\geq 2 \sum_{k=1}^{N(h)} \sqrt{1 + \left|\frac{\nabla u_h^k - \nabla v_h^k}{2}\right|^2} \\ &= 2N(h) \sum_{k=1}^{N(h)} \frac{1}{N(h)} \sqrt{1 + \left|\frac{\nabla u_h^k - \nabla v_h^k}{2}\right|^2} \\ &\geq 2N(h) \sqrt{1 + \left|\sum_{k=1}^{N(h)} \frac{1}{N(h)} \frac{\nabla u_h^k - \nabla v_h^k}{2}\right|^2} \\ &= \sqrt{4N(h)^2 + |\nabla m|^2}. \end{aligned}$$

Therefore, by (8) and (10),

$$P(E^s) \leq \sum_{k=1}^{N(h)} \int_{G_h} \sqrt{4N(h)^2 + |\nabla m|^2} \mathrm{d}z \leq P(E). \tag{11}$$

Let us denote

$$D \coloneqq \{z \in \mathbb{R}^{n-1} \colon E_z \text{ is not an interval}\} = \bigcup \{G_h \colon N(h) \geq 2\}.$$

We have just found that

$$\begin{aligned} P(E) - P(E^s) &\geq \sum_{h=1}^{M} \int_{G_h \cap D} \left(\sqrt{4N(h)^2 + |\nabla m|^2} - \sqrt{4 + |\nabla m|^2} \right) \mathrm{d}z \\ &= \sum_{h=1}^{M} \int_{G_h \cap D} \frac{4(N(h)^2 - 1)}{\sqrt{4N(h)^2 + |\nabla m|^2} + \sqrt{4 + |\nabla m|^2}} \mathrm{d}z \\ &\geq 2 \frac{1}{\sqrt{4N(h)^2 + |\nabla m|^2}} \mathrm{d}z. \end{aligned}$$

Thus, by Cauchy-Schwarz inequality and (11),

$$\begin{aligned} 2\mathscr{L}^{n-1}(D)^2 &= 2 \left(\int_D \frac{(4N(h)^2 + |\nabla m|^2)^{1/4}}{(4N(h)^2 + |\nabla m|^2)^{1/4}} \right)^2 \\ &\geq P(E)(P(E) - P(E^s)). \end{aligned} \tag{12}$$

Step 2. Now let $E \subset \mathbb{R}^n$ with $|E| < \infty$ and $P(E) < \infty$. By Theorem 2.4 there exists a sequence of bounded sets E_h with polyhedral boundary such that

$$E_h \to E, \qquad P(E_h) \to P(E).$$

Note that we can assume that the outer unit normal of ∂E_h is never orthogonal to e_n. Indeed, this is an "open condition", so if it happens that the unit normal of ∂E_h is orthogonal to e_n, we only need to rotate slightly ∂E. Doing smaller and smaller rotations as $h \to \infty$, we still have that $E_h \to E$ (while $P(E_h) \to P(E)$ isn't affected by rotations).

For every $h \in \mathbb{N}$, denote

$$\begin{aligned} m_h(z) &\coloneqq \mathscr{L}^1((E_h)_z), \quad m(z) \coloneqq \mathscr{L}^1(E_z) \qquad \forall z \in \mathbb{R}^{n-1} \\ G_h &\coloneqq \{z \in \mathbb{R}^{n-1} \colon m_h(z) > 0\}, \qquad G \coloneqq \{z \in \mathbb{R}^{n-1} \colon m(z) > 0\} \\ D_h &\coloneqq \{z \in \mathbb{R}^{n-1} \colon (E_h)_z \text{ is not an interval}\}. \end{aligned}$$

Applying step 1 to each E_h, by (11) and (12) we get

$$P(E_h^s) \leq P(E_h^s), \qquad 2\mathscr{L}^{n-1}(D_h) \leq P(E_h)(P(E_h) - P(E_h^s)). \tag{13}$$

Moreover, by Fubini's theorem,

$$\begin{aligned}|E\triangle E_h| &= \int_{\mathbb{R}^{n-1}} \mathscr{L}^1((E_h)_z\triangle E_z)\mathrm{d}z \\ &\geq \int_{\mathbb{R}^{n-1}} |m_h(z)-m(z)|\mathrm{d}z = |E_h^s\triangle E^s|,\end{aligned}$$

so in particular $E_h^s \to E^s$ and by lower semicontinuity

$$P(E^s) \leq \liminf_{h\to\infty} P(E_h^s) \leq \lim_{h\to\infty} P(E_h) = P(E).$$

Moreover,

$$G_h \to G, \qquad (E_h)_z \to E_z \quad \text{for a.e. } z\in\mathbb{R}^{n-1} \tag{14}$$

By Proposition 3.1 the vertical z-slices $(E_h)_z$ and E_z have finite perimeter for a.e. $z\in\mathbb{R}^{n-1}$, thus by the lower semicontinuity of the perimeter

$$P(E_z) \leq \liminf_{h\to\infty} P((E_h)_z). \tag{15}$$

Now, assume that $P(E) = P(E^s)$. Then, by (13), $\chi_{D_h} \to 0$ in $L^1(\mathbb{R}^{n-1})$, so, up to passing to a subsequence, we have the pointwise limit

$$\chi_G(z) = \lim_{h\to\infty} \chi_{G_h\setminus D_h}(z)$$

for a.e. $z\in\mathbb{R}^{n-1}$. By (15),

$$\chi_G(z)P(E_z) \leq \liminf \chi_{G_h\setminus D_h}(z)P((E_h)_z) \qquad \text{for a.e. } z\in\mathbb{R}^{n-1},$$

therefore, by Fatou's lemma,

$$\begin{aligned}\int_G P(E_z)\mathrm{d}z &\leq \int_{\mathbb{R}^n} \liminf \chi_{G_h\setminus D_h}(z)P((E_h)_z)\mathrm{d}z \\ &\leq \liminf_{h\to\infty} \int_{G_h\setminus D_h} P((E_h)_z)\mathrm{d}z \\ &= 2\liminf_{h\to\infty} \mathscr{L}^{n-1}(G_h\setminus D_h) \\ &= 2\mathscr{L}^{n-1}(G) \\ &\leq \int_G P(E_z)\mathrm{d}z\end{aligned}$$

having used the 1-dimensional isoperimetric inequality in the last line, i.e., that $P(E_z)\geq 2$ for a.e. $z\in\mathbb{R}^{n-1}$. This implies that $P(E_z)=2$ for a.e. $z\in\mathbb{R}^{n-1}$, so, by the characterization of the equality in the 1-dimensional case, E_z is an interval for a.e. $z\in\mathbb{R}^{n-1}$.

Step 3. Assume that E is convex, $|E|<\infty$ and $P(E)=P(E^s)$. Since $\mathscr{L}^n(\partial E)=0$[1], we may assume that E is open. Then the projection $G\subset\mathbb{R}^{n-1}$

[1] Indeed, without loss of generality assume that $0\in\mathring{C}$. Then by convexity, for every $\varepsilon>0$

$$\partial C \subset \left(\frac{1}{1-\varepsilon}\mathring{C}\right)\setminus C.$$

is a convex set, and for every $z \in G$, the vertical z-slice is

$$E_z = (-v(z), u(z)),$$

where $v, u\colon G \to \mathbb{R}$ are respectively convex and concave functions, so in particular they are locally Lipschitz and E has Lipschitz boundary. By Rademacher's theorem, ∇u and ∇v are defined at a.e. $z \in G$, so

$$\begin{aligned} P(E) &= \int_G \left(\sqrt{1+|\nabla u|^2} + \sqrt{1+|\nabla v|^2}\right) \mathrm{d}z \\ P(E^s) &= \int_G \sqrt{1 + \left|\frac{\nabla u + \nabla v}{2}\right|^2} \mathrm{d}z \end{aligned}$$

since

$$E^s = \left\{(z,t) \in G \times \mathbb{R}\colon |t| < \frac{u(z)+v(z)}{2}\right\}.$$

Since the function $z \mapsto \sqrt{1+|z|^2}$ is strictly convex,

$$\sqrt{1+|\nabla u|^2} + \sqrt{1+|\nabla v|^2} \geq \sqrt{1 + \left|\frac{\nabla u + \nabla v}{2}\right|^2}$$

with equality if and only if $\nabla u = \nabla v$. Hence, if $P(E^s) = P(E)$, the equality holds almost everywhere and there must be that $\nabla u = \nabla v$ a.e. in G. Thus, there exists $c \in \mathbb{R}$ such that $u(z) - v(z) = c$ for a.e. $z \in G$, which implies that

$$\begin{aligned} E &= \{(z,t) \in G \times \mathbb{R}\colon -u(z) - c < t < u(z)\}, \\ E^s &= \{(z,t) \in G \times \mathbb{R}\colon -u(z) - c/2 < t < u(z) + c/2\}, \end{aligned}$$

that is $E = E^s + \frac{c}{2} e_n$. □

Recall that the density of E at a point $x \in \mathbb{R}^n$ is

$$\Theta(E,x) := \lim_{r \to 0} \frac{|E \cap B_r(x)|}{\omega_n r^n}$$

whenever it exists. It's easy to show that, whenever this limit exists, it's actually equal to

$$\Theta(E,x) = \lim_{r \to 0} \frac{|E \cap Q^n(x,r)|}{2^n r^n} \tag{16}$$

where

$$Q^n(x,r) = x + (-r,r)^n$$

is the cube centered at x with side $2r$.

We denote

$$E^{(1)} := \{x \in \mathbb{R}^n\colon \Theta(E,x) = 1\}$$

the set of with density one points of E. A standard application of Besicovitch's covering theorem shows that $|E \triangle E^{(1)}| = 0$.

Lemma 3.3. *Let $E \subset \mathbb{R}^n$ be a set of locally finite perimeter and suppose that E_z is equivalent to an interval for a.e. $z \in \mathbb{R}^{n-1}$. Then $(E^{(1)})_z$ is an interval for every $z \in \mathbb{R}^{n-1}$.*

Proof. Let $z_0 \in \mathbb{R}^{n-1}$, and suppose that $x_1 = (z_0, t_1), x_2 = (z_0, t_2) \in E^{(1)}$ for some $t_1 < t_2$, and fix $t_0 \in (a, b)$. We have to prove that $x_0 = (z_0, t_0) \in E^{(1)}$, that is $\Theta(E, x_0) = 1$.

Fix $0 < \varepsilon < 1$, and for every $t \in \mathbb{R}$, $r > 0$ denote $I_r(t) = B_r(t) = (t-r, t+r)$. Since $x_1, x_2 \in E^{(1)}$, $\Theta(E, x) = \Theta(E, y) = 1$, so by (16) there exists $\overline{r} > 0$ such that for every $r < \overline{r}$

$$|E \cap Q^n(x_k, r)| \geq \left(1 - \frac{\varepsilon}{2}\right) 2^n r^n, \quad I_r(t_0) \cap I_r(t_k) = \varnothing \qquad k = 1, 2. \tag{17}$$

Fix $r < \overline{r}$ and for $k = 1, 2$ denote by G_k the set of $z \in Q^{n-1}(z_0, r)$ such that E_z is equivalent to an interval and $\mathscr{L}^1(E_z \cap I_r(t_k)) > 0$. Since E_z is equivalent to an interval for a.e. $z \in \mathbb{R}^{n-1}$, by Fubini's theorem

$$|E \cap Q^n(x_k, r)| = \int_{G_k} \mathscr{L}^1(E_z \cap I_r(t_k)) \mathrm{d}z \leq 2r \mathscr{L}^{n-1}(G_k).$$

Combining this with (17),

$$\mathscr{L}^{n-1}(G_k) \geq \left(1 - \frac{\varepsilon}{2}\right) 2^{n-1} r^{n-1}, \qquad k = 1, 2.$$

Therefore,

$$\begin{aligned}
\mathscr{L}^{n-1}(G_1 \cap G_2) &= \mathscr{L}^{n-1}(G_1) + \mathscr{L}^{n-1}(G_2) - \mathscr{L}^{n-1}(G_1 \cup G_2) \\
&\geq \mathscr{L}^{n-1}(G_1) + \mathscr{L}^{n-1}(G_2) - \mathscr{L}^{n-1}(Q^{n-1}(z_0, r)) \\
&\geq (1 - \varepsilon) 2^{n-1} r^{n-1}.
\end{aligned}$$

Now let $z \in G_1 \cap G_2$. Since E_z is equivalent to an interval and $\mathscr{L}^1(E_z \cap I_r(t_k)) > 0$ for $k = 1, 2$, by (17) we have

$$\mathscr{L}^1(E_z \cap I_r(t_0)) = 2r.$$

Integrating this over $G_1 \cap G_2$, we find

$$\begin{aligned}
|E \cap Q^n(x_0, r)| &\geq \int_{G_1 \cap G_2} \mathscr{L}^1(E_z \cap I_r(t_0)) \mathrm{d}z \\
&= 2r \mathscr{L}^{n-1}(G_1 \cap G_2) \geq (1 - \varepsilon) 2^n r^n.
\end{aligned}$$

for every $r < \overline{r}$. Since $\varepsilon > 0$ was arbitrary,

$$\lim_{r \to 0} \frac{|E \cap Q^n(x_0, r)|}{2^n r^n} = 1,$$

which proves that $x_0 \in E^{(1)}$. □

4 The isoperimetric inequality

Lemma 4.1 (Constrained isoperimetric inequality)**.** *Let $m, R > 0$ with $m < \omega_n R^n$. Then*

$$\alpha(m,R) := \inf\{P(E)\colon E \subset B_R \text{ measurable, } |E| = m\}$$

is realized, and every minimizer is equivalent to a ball. In particular, $\alpha(m,R)$ is the perimeter of a ball of volume m, that is

$$\alpha(m,R) = n\omega_n^{1/n} m^{\frac{n-1}{n}}.$$

Proof. That $\alpha(m,R)$ is realized is a consequence of the Direct Method, as bounded sets in $BV(B_R)$ are precompact with respect to the BV-strict convergence, the family of characteristic functions of sets in B_R is closed, and any limit in the class preserves the volume.

Let us now assume that $E \subset B_R$ satisfies $|E| = m$ and $P(E) = \alpha(m,R)$. Since $|E \triangle E^{(1)}| = 0$, we may assume that $E = E^{(1)}$. For every unit vector $v \in \mathbb{R}^n$, $|v| = 1$, by Theorem 3.2 we have that $P(S_v(E)) \le P(E)$. On the other hand, by the properties of the Steiner symmetrization $|S_v(E)| = m$ and $\operatorname{diam}(S_v(E)) \le \operatorname{diam} E < R$, so $P(S_v(E)) \ge \alpha(m,R) = P(E)$. Then by Theorem 3.2 and Lemma 3.3

$$E_{v,z} := \{s \in \mathbb{R}\colon z + sv \in E\}$$

is an interval for every $z \in v^\perp$.

We claim that E is convex. Indeed, let $x, y \in E$ be two distinct points, and set $v = (y-x)/|y-x|$. Then if $z \in v^\perp$ is the projection of x (and y) in $v^\perp$,

$$(1-\lambda)x + \lambda y = z + [(1-\lambda)x \cdot v + \lambda x \cdot v]v \in E, \qquad \forall \lambda \in [0,1]$$

as $[x \cdot v, y \cdot v] \subset E_{v,z}$.

By Theorem 3.2, for every unit vector $v \in \mathbb{R}^n$ there exists $c_v \in \mathbb{R}$ such that $E = S_v(E) + c_v v$. Consider

$$E' := E - (c_{e_1} e_1 + \cdots c_{e_n} e_n).$$

Since E' is still convex and for every unit vector $v \in \mathbb{R}^n$ we still have $P(E') = P(S_v(E'))$, there are constants $c'_v \in \mathbb{R}$ such that $E' = S_v(E') + c'_v v$. But by construction,

$$S_{e_i}(E') = S_{e_i}(E) - \sum_{j \ne i} c_j e_j = E - c_i e_i - \sum_{j \ne i} c_j e_j = E',$$

so $c'_i = 0$ for every $i = 1, \dots, n$. Hence, E' is invariant under the reflections with respect to the coordinates planes, and therefore invariant under the map $x \mapsto -x$ (the composition of all those reflections). Since for every unit vector v we can write $E' = S_v(E) + c'_v v$, we have

$$S_v(E) + c'_v v = -S_v(E) - c'_v v = S_v(E) + c'_v(-v). \tag{18}$$

Clearly, $c'_{-v} = -c'_v$ for every unit vector v. On the other hand, combining (18) with the fact that $S_v(E) = S_{v'}(E)$, we also have that $c'_v = c'_{-v}$. Therefore, for every unit vector $v \in \mathbb{R}^n$, it must be $c'_v = 0$, that is, $S_v(E') = E'$.

As a consequence, E' is invariant under reflections across hyperplanes through the origin. Since compositions of such reflections generate all rotations fixing the origin, it follows that E' is invariant under every rotation that fixes the origin. Hence, all the sections

$$E'_v := \{s \in \mathbb{R} \colon sv \in E'\}, \qquad v \in \mathbb{R}^n,\ |v| = 1,$$

must be equal, and since E' is bounded there exists $r > 0$ such that $E'_v = (-r, r)$ for every unit vector $v \in \mathbb{R}^n$. Hence, $E' = B_r(0)$ and $E = B_r(p)$, with $p = (c_{e_1}, \dots, c_{e_n})$. □

We can now prove the isoperimetric inequality.

Proof of Theorem 1.1. Suppose that E is bounded. Then taking $R > 0$ big enough, by Lemma 4.1,

$$P(E) \geq n\omega_n^{1/n}|E|^{\frac{n-1}{n}},$$

and the equality holds if and only if E is equivalent to a ball.

If E is not bounded, by Theorem 2.4 we can approximate it with bounded sets, that is, there are bounded sets $E_h \subset \mathbb{R}^n$ with $E_h \to E$ and $P(E_h) \to P(E)$. Passing to the limit, the inequality is preserved.

Now assume that $E = E^{(1)}$ is unbounded and $|E| < \infty$. Assume by contradiction that

$$P(E) = n\omega_n^{1/n}|E|^{\frac{n-1}{n}}.$$

By (3.2), $P(E) = P(S_v(E))$ for every unit vector $v \in \mathbb{R}^n$. Up to a translation, arguing exactly as in Lemma 4.1 we have that all the sections E

$$E_v := \{s \in \mathbb{R} \colon sv \in E\}, \qquad v \in \mathbb{R}^n,\ |v| = 1,$$

are equal. Since E is unbounded, we have $E_v = \mathbb{R}$ for every unit vector $v \in \mathbb{R}^n$, but then $E = \mathbb{R}^n$, a contradiction in because we assumed that $|E| < \infty$. □

5 An alternative approach

Adopting the point of view of differential geometry, we could have used a completely different approach. Indeed, fix $m > 0$ and let

$$\alpha(m) := \inf \left\{ \mathscr{H}^{n-1}(\Sigma) \;\middle|\; \begin{array}{l} \Sigma = \partial\Omega \text{ smooth, connected,} \\ \text{compact hypersurface}, |\Omega| = m \end{array} \right\}. \tag{19}$$

Suppose that $\Sigma = \partial\Omega$ is a minimizer, that is, $\mathscr{H}^{n-1}(\Sigma) = \alpha(m)$ and $|\Omega| = m$. Let $\nu \colon \Sigma \to \mathbb{S}^{n-1}$ be the outer unit normal of Σ, A its second fundamental form,

defined[2] by

$$A(X,Y) = \mathrm{d}\nu_x[X] \cdot Y$$

for every tangent vectors $X, Y \in T_x\Sigma$, and $H = \mathrm{tr}(A)$ its mean curvature. By the inverse function theorem and by the compactness of Σ, there is $\varepsilon > 0$ such that

$$\phi(t,x) = x + t\nu(x),$$

is a diffeomorphism on $(-\varepsilon, \varepsilon) \times \Sigma$. Let $u\colon \Sigma \to \mathbb{R}$ be a smooth function with zero average, i.e., $\int_\Sigma u \mathrm{d}\mathscr{H}^{n-1} = 0$, and define

$$F(t,x) = \phi(tu(x), x)$$

for every $x \in \Sigma$ and $t \in \mathbb{R}$ small enough. Denote

$$\Sigma_t = \{F(t,x)\colon x \in \Sigma\}.$$

Then, $\Sigma_t = \partial\Omega_t$ is the boundary of a bounded set Ω_t. For every $x \in \Sigma$, consider the geodesic

$$\gamma_x(s) = \phi(s,x),$$

which has constant velocity $|\dot\gamma_x| = u(x)$. Then, by the coarea formula,

$$\begin{aligned}
|\Omega_t| &= |\Omega| + |\{\phi(s,x)\colon 0 < s < tu(x)\}| - |\{\phi(s,x)\colon tu(x) < s < 0\}| \\
&= m + \int_{\{u>0\}} \int_0^t |\dot\gamma_x| \mathrm{d}t \mathrm{d}\mathscr{H}^{n-1}(x) - \int_{\{u<0\}} \int_0^t |\dot\gamma_x| \mathrm{d}t \mathrm{d}\mathscr{H}^{n-1}(x) \\
&= m + t\int_\Sigma u \ \mathrm{d}\mathscr{H}^{n-1} = m,
\end{aligned}$$

hence Σ_t is a competitor in (19) for every t. By the minimality of Σ,

$$\mathscr{H}^{n-1}(\Sigma_t) \geq \mathscr{H}^{n-1}(\Sigma),$$

so the function $t \mapsto \mathscr{H}^{n-1}(\Sigma_t)$ has a minimum at $t = 0$. The first variation formula implies that

$$0 = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \mathscr{H}^{n-1}(\Sigma_t) = \int_\Sigma Hu \ \mathrm{d}\mathscr{H}^{n-1}.$$

Since this hold for every $u : \Sigma \to \mathbb{R}$ with zero average, H is constant, and we say that Σ is *CMC*.

Taking the second derivative with respect to t, the second variation formula implies that

$$\begin{aligned}
0 &\leq \int_\Sigma |\nabla_\Sigma u|^2 - (|A|^2 - H^2)u^2 \ \mathrm{d}\mathscr{H}^{n-1} \\
&= \int_\Sigma |\nabla_\Sigma u|^2 - |A|^2 u^2 \ \mathrm{d}\mathscr{H}^{n-1} = -\int_\Sigma u L_\Sigma u \ \mathrm{d}\mathscr{H}^{n-1}
\end{aligned} \tag{20}$$

[2]Note that, identifying the tangent spaces as subspaces of $\mathbb{R}^n$ in the usual way, the scalar product makes sense. Actually, $T_{\nu(x)}\mathbb{S}^{n-1} = T_x\Sigma \subset \mathbb{R}^n$ by definition of unit normal.

for every smooth function u with zero average, where

$$L_\Sigma u = \Delta_\Sigma u + |A|^2 u$$

is the *Jacobi operator* of Σ. Any hypersurface Σ that satisfies (20) is called *CMC-stable*.

This approach can give another proof of the isoperimetric inequality. In particular, it holds the following stronger result.

Theorem 5.1 (Barbosa-do Carmo, [BC84])**.** *Let Σ be a compact, connected, orientable, CMC hypersurface in $\mathbb{R}^n$. Then Σ is CMC-stable if and only if Σ is a round sphere.*

Proof. Let Σ be a sphere, and assume without loss of generality that $\Sigma = \mathbb{S}^{n-1}$ has radius 1 and is centered at the origin. Since $\nu(x) = x$, we have $\mathrm{d}\nu = \mathrm{id}$, hence $H = (n-1)$ and $|A|^2 = (n-1)$. Thus, $\mathbb{S}^{n-1}$ is CMC. The spectrum of the spherical Laplacian $\Delta_{\mathbb{S}^{n-1}}$ is well known thanks to the formula of the standard Laplacian of $\mathbb{R}^n$ in spherical coordinates

$$\Delta_{\mathbb{R}^n} = \frac{\partial^2}{\partial r^2} + \frac{n-1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\Delta_{\mathbb{S}^{n-1}},$$

that allows us to identify the homogeneous harmonic functions of $\mathbb{R}^n$ with the eigenfunctions of $\Delta_{\mathbb{S}^{n-1}}$. In particular, the first nonzero eigenvalue of $\Delta_{\mathbb{S}^{n-1}}$ is

$$\begin{aligned}\lambda_1 &= \inf\left\{\frac{\int_{\mathbb{S}^{n-1}} |\nabla_{\mathbb{S}^{n-1}} u|^2 \mathrm{d}\mathscr{H}^{n-1}}{\int_{\mathbb{S}^{n-1}} u^2 \mathrm{d}\mathscr{H}^{n-1}} : u \in C^\infty(\mathbb{S}^{n-1}) \setminus \{0\},\ \int_{\mathbb{S}^2} u \ \mathrm{d}\mathscr{H}^{n-1} = 0\right\} \\ &= (n-1).\end{aligned}$$

Thus, for every smooth function $u\colon \mathbb{S}^{n-1} \to \mathbb{R}$ with zero average

$$\int_{\mathbb{S}^{n-1}} |\nabla_{\mathbb{S}^{n-1}} u|^2 - (n-1)u^2 \ \mathrm{d}\mathscr{H}^{n-1} \geq 0,$$

that is, $\mathbb{S}^{n-1}$ is CMC-stable.

Now assume that Σ is CMC-stable, and let us prove that it must be a sphere. Consider the function $f\colon \Sigma \to \mathbb{R}$ defined by

$$f(x) = x \cdot \nu(x)$$

and the radial vector field $X(x) = x$, for every $x \in \Sigma$. Even if X is not tangent, we can define its divergence on Σ as

$$\operatorname{div}_\Sigma X = \sum_{i=1}^{n-1} (D_{\tau_i} X) \cdot \tau_i$$

locally for any orthonormal frame $\tau_1, \dots, \tau_n$ of Σ. Since X is the restriction of the identity,

$$\operatorname{div}_\Sigma X = \sum_{i=1}^{n-1} (D_{\tau_i} X) \cdot \tau_i = \sum_{i=1}^{n-1} \tau_i \cdot \tau_i = n-1.$$

By the generalized divergence theorem for submanifolds,

$$\begin{aligned}\int_\Sigma \operatorname{div}_\Sigma X \,\mathrm{d}\mathscr{H}^{n-1} &= \int_{\partial\Sigma} X\cdot\nu_{\partial\Sigma}\,\mathrm{d}\mathscr{H}^{n-2} + \int_\Sigma HX\cdot\nu\,\mathrm{d}\mathscr{H}^{n-1}\\ &= H\int_\Sigma f(x)\,\mathrm{d}\mathscr{H}^{n-1}(x)\end{aligned}$$

because $\partial\Sigma=\varnothing$. Therefore, $u=Hf-n+1$ has zero average, and can be tested in (20).

With a clever choice of frame, one can compute that

$$\Delta_\Sigma f = H - |A|^2 f. \tag{21}$$

Therefore,

$$L_\Sigma u = H^2 - |A|^2 Hf + |A|^2(Hf-n+1) = H^2-(n-1)|A|^2,$$

and (20) becomes

$$\begin{aligned}0 &\le \int_\Sigma ((n-1)|A|^2 - H^2)(Hf-n+1)\,\mathrm{d}\mathscr{H}^{n-1}\\ &= (n-1)\int_\Sigma |A|^2(Hf-n+1)\,\mathrm{d}\mathscr{H}^{n-1}.\end{aligned}$$

By the divergence theorem and by (21)

$$0=\int_\Sigma \Delta_\Sigma f\,\mathrm{d}\mathscr{H}^{n-1} = H\mathscr{H}^{n-1}(\Sigma) - \int_\Sigma |A|^2 f\,\mathrm{d}\mathscr{H}^{n-1},$$

so

$$0\le\int_\Sigma |A|^2(Hf-n+1)\,\mathrm{d}\mathscr{H}^{n-1} = \int_\Sigma H^2-(n-1)|A|^2\mathrm{d}\mathscr{H}^{n-1},$$

that is

$$(n-1)\int_\Sigma |A|^2\mathrm{d}\mathscr{H}^{n-1} \le \int_\Sigma H^2\mathrm{d}\mathscr{H}^{n-1}. \tag{22}$$

However, if $\lambda_1,\dots,\lambda_{n-1}$ are the eigenvalues of A, by the $AM\le QM$ inequality we have that

$$H^2 = \left(\sum_{i=1}^{n-1}\lambda_i\right)^2 \le (n-1)\sum_{i=1}^{n-1}\lambda_i^2 = (n-1)|A|^2,$$

with equality if and only if $\lambda_1=\cdots=\lambda_{n-1}$. But by (22), it must be $H^2=(n-1)|A|^2$, that is,

$$\mathrm{d}\nu = \frac{H}{n-1}\mathrm{id}$$

(in this case, Σ is called *totally umbilical*). This implies that Σ is a sphere. Indeed, consider the map $F\colon\Sigma\to\mathbb{R}^n$ given by

$$F(x) = \frac{H}{n-1}x - \nu(x)$$

for every $x \in \Sigma$. Its differential vanishes, so, since Σ is connected, it's constant: there exists $p \in \mathbb{R}^n$ such that

$$F(x) = \frac{H}{n-1} p$$

for every $x \in \Sigma$. But then

$$\nu(x) = \frac{H}{n-1}(x-p),$$

and therefore

$$|x-p| = \frac{n-1}{|H|},$$

for every $x \in \Sigma$, that is, Σ is the sphere of radius $(n-1)/|H|$ centered at p. $\square$

Remark 5.2. *A posteriori*, assuming that Σ is connected in the isoperimetric problem (19) is not restrictive. In fact assume that $\Sigma = \partial\Omega$ is not connected. Let $\Omega_1, \dots, \Omega_k$ be the connected components of Ω, and $\Sigma_0^j, \dots, \Sigma_{N(j)}^j$ the connected components of Σ such that $\partial\Omega_j = \bigcup_i \Sigma_i^j$. Since each Σ_i^j is CMC-stable, by Theorem 5.1 it's a round sphere, so Σ_i^j a boundary of a ball B_i^j. Hence, every Ω_j must be obtained by removing some disjoint balls $B_1^j, \dots, B_{N(j)}^i$ from an initial big ball B_0^j of radius R_j. Denote $\Omega^* = \bigcup_j B_0^j \supset \Omega$ and $\Sigma^* = \partial\Omega^* \subset \Sigma$. Then it's easy to see (for example, by induction over k) that

$$\begin{aligned}
\mathscr{H}^{n-1}(\Sigma^*)^n &= \left(n\omega_n \sum_{j=1}^k R_j^{n-1}\right)^n = n^n\omega_n^n \left(\sum_{j=1}^k R_j^{n-1}\right)^n \\
&\geq n^n\omega_n^n \left(\sum_{j=1}^k R_j^n\right)^{n-1} = n^n\omega_n^{n-1}\left(\omega_n \sum_{j=1}^k R_j^n\right)^{n-1} \\
&= n^n\omega_n|\Omega^*|^{n-1}
\end{aligned}$$

with strict inequality whenever $k > 1$, which is indeed the case if $\Sigma = \Sigma^*$. If instead $\Sigma \neq \Sigma^*$, then observe that

$$\mathscr{H}^{n-1}(\Sigma) > \mathscr{H}^{n-1}(\Sigma^*) \geq n\omega_n^{1/n}|\Omega^*|^{\frac{n-1}{n}} > n\omega_n^{1/n}|\Omega|^{\frac{n-1}{n}}.$$

References

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